

A Steady-State Throughput Analysis of Cluster Tools: Dual-Blade Versus Single-Blade Robots

Srilakshmi Venkatesh, *Associate Member, IEEE*, Rob Davenport, Pattie Foxhoven,
and Jaim Nulman, *Senior Member, IEEE*

Abstract—An analysis of throughput in a cluster tool with a dual-blade robot operating in steady-state mode is presented. The analysis is based on a single-wafer serial processing cluster tool. Two types of schedules are distinguished, called *transport-bound* schedules and *process-bound* schedules. In a transport-bound schedule changes in process times do not affect the throughput of the cluster tool, and denotes the maximum throughput achievable in the cluster tool. In a process-bound schedule, the process time predominates the effect on the throughput. The analysis indicates that a dual-blade robot improves the throughput of the cluster tool over a single-blade robot under process-bound conditions. Under process-bound conditions, a cluster tool with a single-blade robot would need to double the speed of the robot, compared with a dual-blade robot of equivalent speed, to achieve similar throughput. Under transport-bound conditions, the throughput of the cluster tool is the same for both dual-blade and single-blade cluster tools.

Index Terms—Flexible manufacturing system, flow-line automation, scheduling/sequencing.

I. INTRODUCTION

RECENT trends show an increase in the use of cluster tools in semiconductor fabrication facilities [1], [2]. A cluster tool consists of a number of process chambers and loadlock, all accessible through a central chamber that houses a robot for transferring wafers (Fig. 1). The loadlock is used to load and unload batches of wafers. Existing literature is aimed at understanding the performance of cluster tools [3]–[8]. *Throughput* has been one of the common measures that have been used to understand the behavior of cluster tools. Throughput is defined as the processing rate of the cluster tool, usually expressed in wafers per hour.

A distinction is made between two types of robots for transferring wafers in a cluster tool, referred to as *single-blade* and *dual-blade* robots. The single-blade robot can carry one wafer at a time. In the case of cluster tools with a single-blade robot, throughput studies are presented in Perkinson *et al.* [5] and Perkinson, Gyurcsik, and McLarty [6]. The dual-blade robot consists of two blades pointing in opposite directions each capable of carrying a wafer (Fig. 1). The two blades are *tightly-coupled* by construction, i.e., at anytime only one of the two blades can pick-up or drop-off a wafer to or from a chamber. In this paper, expressions to calculate the steady

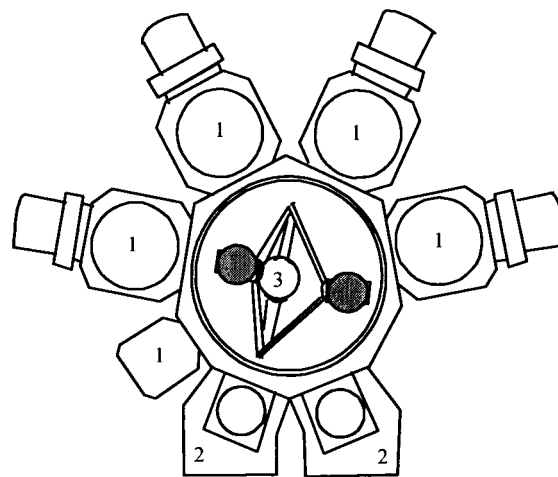


Fig. 1. An example five-chamber cluster tool with a dual-blade robot. 1: chambers, 2: loadlocks, and 3: a dual-blade robot with a wafer on each blade.

state throughput for a single-wafer serial-processing cluster tool with a dual-blade robot are presented. This is followed by a comparison of performance of a dual-blade cluster tool with a single-blade cluster tool. Note that throughout this document the terms dual-blade and single-blade are used to qualify both cluster tools and robots, interchangeably.

Steady-state throughput of the cluster tool is characterized by the *Fundamental Period* (FP) introduced in Perkinson *et al.* [5]. FP is defined as the elapsed time between the completion of processing of two consecutive wafers [5]. The steady-state throughput of the cluster tool can be calculated by taking the inverse of the FP [8]. Informally, a cluster tool is in steady-state when the flow of wafers in and out of the cluster tool is the same [5]. The cluster tool is said to be in a transient-state during the initiating and terminating of the processing of a batch of wafers [5]. Note that in cluster tools with multiple loadlocks, the cluster tool can be in steady-state for long periods of time by having different loadlocks be alternately ready for processing. This can be achieved by readying (i.e., pumping down to chamber pressures, etc.) one of the loadlocks to send wafers into the cluster tool, while wafers in the other loadlock is currently being processed. In this paper, the throughput effect of transient-state is ignored.

II. ASSUMPTIONS AND PRELIMINARIES

This study deals with an N chamber cluster tool excluding the loadlocks. Note that the cluster tool can have more than one

Manuscript received November 11, 1996; revised June 16, 1997.
The authors are with Metal Deposition Products Group, Applied Materials,
Santa Clara, CA 95054 USA (e-mail: venkatesh_s@amat.com).
Publisher Item Identifier S 0894-6507(97)07551-9.

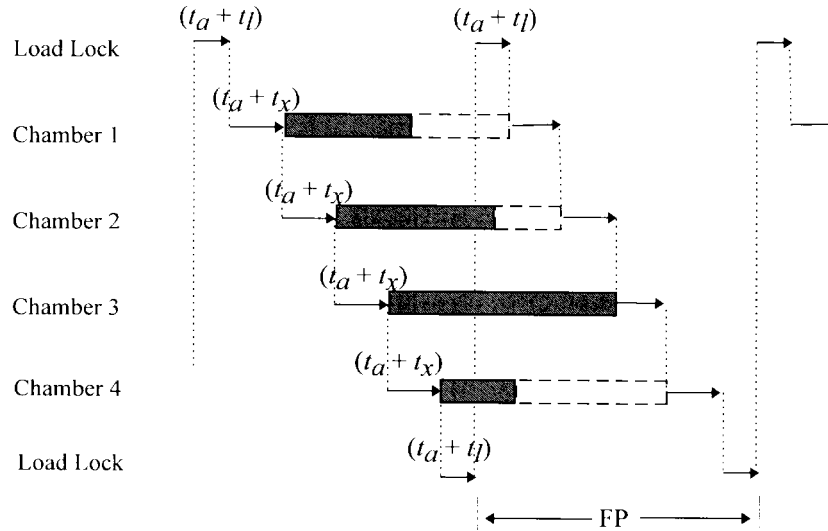


Fig. 2. A steady-state timing diagram of a transport-bound schedule for a four-chamber cluster tool with a dual-blade robot.

loadlock. The analysis presented in this paper assumes that the loadlock does not constrain the cluster tool throughput. Wood [8] presents a derivation of the number of loadlocks required to achieve this condition.

In the current analysis, it is assumed that a wafer w , beginning in the loadlock, is processed through all the chambers c_i , $i = 1, 2, \dots, N$, before being returned to the loadlock. For simplicity, wafer w is assumed to be processed sequentially through chambers c_i , $i = 1, 2, \dots, N$. Thus, chamber c_1 is the first processing step and chamber c_N is the last processing step. The processing time in chamber c_i is denoted by p_i . The maximum processing time in the cluster tool is denoted by P , i.e., P is the maximum of all process times p_i , $i = 1, 2, \dots, N$. Individual wafers w will be denoted with subscripts wherever appropriate.

All the wafers in the batch are assumed to be identical and go through the same processing sequence. The process time for each wafer for a particular step (i.e., p_i for some i) is identical. It is assumed that there are no redundant process chambers and, hence, each wafer goes through all the chambers in the cluster tool. Furthermore, each process chamber can accommodate a single wafer at a time. It is assumed that the wafers are moved directly from one chamber to the subsequent chamber, after the completion of the current process.

A dual-blade robot begins and ends the wafer movement with both the blades empty. In steady state, an unprocessed wafer is picked-up from the loadlock on one blade, which is subsequently swapped with another wafer that has finished processing in the next processing step. During the intermediate steps both the blades can be full, with a wafer on each blade, as the wafer-on-the-blade is exchanged with the wafer-in-the-chamber. Also, a wafer may wait on the blade of the robot for another wafer in the subsequent step to complete its processing. Finally, note that at the beginning and the end of the sequence, the robot does not exchange wafers but merely picks-up or drops-off a wafer.

Next, the steps involved in exchanging a wafer-on-the-blade with a wafer-in-the-chamber is detailed. In this case, the

transfer of wafer into a chamber involves: 1) the arrival of an empty blade at the chamber (t_a); 2) picking-up the resident wafer-in-the-chamber (t_l); 3) rotating the robot to point the wafer-on-the-blade (from the previous processing step) at the chamber (t_r); and 4) finally, dropping-off the wafer from previous step into the chamber (t_l). The time to exchange the wafer-on-the-blade and the wafer-in-the-chamber, steps (2)–(4) above, is denoted by t_x . Here, it is assumed that the time to lower/lift (t_l) a wafer from/to the robot blade is identical.

III. DERIVATION OF FP

In this section, the expressions for the FP of a dual-blade cluster tool are derived. The analysis is developed using an example four-chamber cluster tool, which is latter generalized to an N chamber cluster tool. Each wafer is processed through chambers c_i , $i = 1, 2, 3, 4$, in order. A distinction is made between *transport-bound* and *process-bound* schedules in the case of dual-blade cluster tools introduced in Perkinson *et al.* [5]. In a transport-bound schedule, the robot is *always* busy and, under these conditions, changes in the process time will not have any effect on the FP. In the case of a process-bound schedule, both the transport times and the process times impact the throughput of the cluster tool. First, the FP for a transport-bound schedule is derived, which is then followed by a derivation of a process-bound schedule.

Fig. 2 shows a steady-state *timing diagram* of a transport-bound schedule. Perkinson *et al.* [5] introduced the concept of a timing diagram for single-blade cluster tools. The timing diagram is a Gantt charting of the chamber and robot usage in cluster tools. In this diagram, filled boxes depict chamber busy periods and empty boxes in dashed lines denote allowable process time limits. Changes in the process time within these time limits does not alter the FP of the cluster tool. The arrows indicate robot movements, and each arrow represents the time required to move a wafer to a chamber. The arrow is

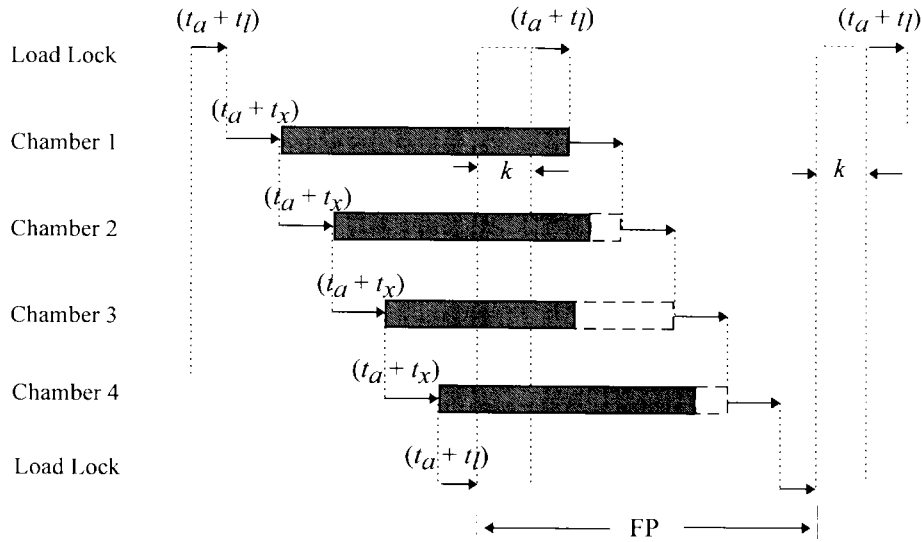


Fig. 3. A steady-state timing diagram of a process-bound schedule for a four-chamber cluster tool with a dual-blade robot; process time p_1 violates transport-bound limits by k time-units.

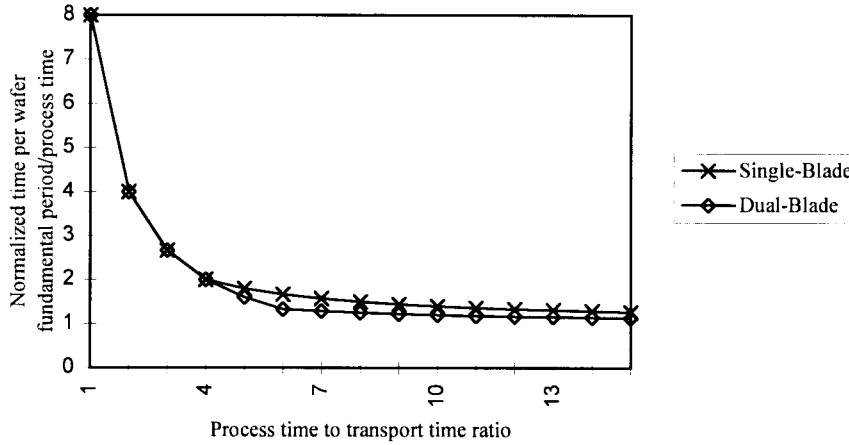


Fig. 4. A comparison of performance with dual and single-blades in a three-chamber cluster tool.

placed in the same row as the chamber the wafer is destined for.

In Fig. 2, each arrow represents: 1) the time for the robot to arrive (t_a) with the wafer to the chamber (or loadlock) from the previous chamber position, and 2) the time to exchange (t_x) with a new wafer with the chamber (or the time to drop-off/pick-up (t_l) a wafer). In the case of loadlocks, only a drop-off or pick-up time (t_l) need to be included. In the case of chambers c_1 , c_2 , c_3 , and c_4 an exchange time (t_x) on the arrow in the timing diagram is included. Note that t_a for picking-up a new wafer, can be regarded as the time required to index a wafer-elevator mechanism to position the wafer for pick-up at the correct vertical level and/or the time required to rotate the robot to point at the appropriate loadlock. This would depend on whether or not the same loadlock is used to pick-up and drop-off a wafer. The analysis presented in this paper does not depend on such nuances in the operation of the cluster tool. From the figure of the transport-bound schedule, one can

derive the FP for the four-chamber cluster tool as follows:

$$FP(\text{transport-bound}) = 6t_a + 4t_x + 2t_l. \quad (1)$$

In a transport-bound schedule, the process times in the different chambers need to obey condition (2). Furthermore, all the process chambers have the same amount of time to complete processing.

$$p_i \leq 5t_a + 3t_x + 2t_l, \quad i = 1, 2, 3, 4. \quad (2)$$

Suppose, now one or more of the process chambers violate the above condition and, consequently, the cluster tool is in a process-bound region (Fig. 3). Let the maximum process time be p_1 . Clearly, if chamber c_1 violates the process time constraint in (3) once, it will violate constraint (3) during each robot cycle by the same amount. This is true because, the process times are assumed to be identical for each wafer for a particular step. From the steady-state timing diagram, it is also clear that each chamber will have p_1 time units to process.

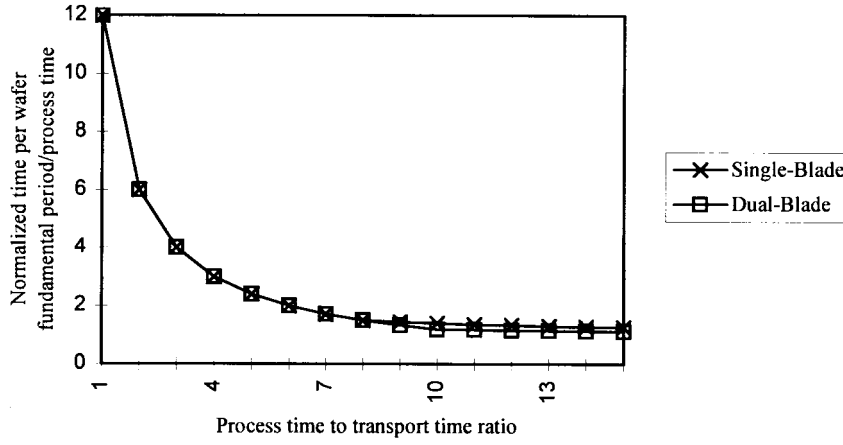


Fig. 5. A comparison of performance with dual and single-blades in a five-chamber cluster tool.

Since p_1 is the maximum process time in the cluster tool, the other process chambers will not further violate (3). Here, the FP can be expressed by the following equation:

$$\text{FP (process-bound)} = p_1 + t_a + t_x. \quad (3)$$

The example analysis can be extended to a general cluster tool with N chambers. The appropriate FP is shown in (4) and (5). The process bounds in a general cluster tool is given in (6).

$$\text{FP (transport-bound)} = (N + 2)t_a + Nt_x + 2t_l \quad (4)$$

$$\text{FP (process-bound)} = P + t_a + t_x \quad (5)$$

$$p_i \leq (N + 1)t_a + (N - 1)t_x + 2t_l, \quad i = 1, 2, \dots, N. \quad (6)$$

IV. PERFORMANCE COMPARISON: DUAL-BLADE VERSUS SINGLE-BLADE

In this section, the performance curves of a cluster tool with a dual-blade robot is presented. A comparison with a single-blade cluster tool is also included. The expression of FP for a single-blade cluster tool is presented in Perkinson *et al.* [5], and reproduced below for convenience. Here, $2T$ is the time required to move a wafer from one chamber to another.

$$\begin{aligned} \text{FP (transport-bound, single-blade)} &= 2T(N + 1), \\ &\text{for } P < 2T(N - 1) \end{aligned} \quad (7)$$

$$\begin{aligned} \text{FP (process-bound, single-blade)} &= P + 4T, \\ &\text{for } P > 2T(N - 1). \end{aligned} \quad (8)$$

Equivalently, in the notation of Perkinson *et al.* [5], the FP of a dual-blade cluster tool in (4) and (5) can be rewritten as (9) and (10), respectively. In developing these expressions to compare the performance between dual-blade and single-blade cluster tools, it is assumed that t_a to be equivalent to t_r and the time required to lift/lower a wafer (t_l) using a dual-blade

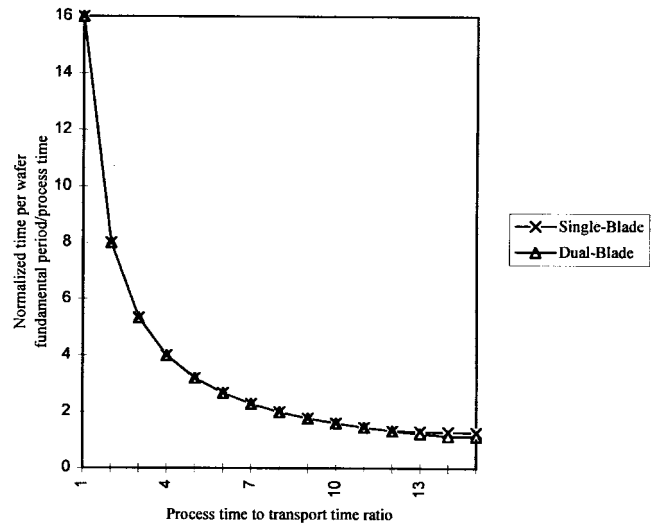


Fig. 6. A comparison of performance with dual and single-blades in a seven-chamber cluster tool.

or a single-blade robot is the same.

$$\begin{aligned} \text{FP (transport-bound, dual-blade)} &= 2T(N + 1), \\ &\text{for } P < 2NT \end{aligned} \quad (9)$$

$$\begin{aligned} \text{FP (process-bound, dual-blade)} &= P + 2T, \\ &\text{for } P > 2NT \end{aligned} \quad (10)$$

$$p_i \leq 2NT, \quad i = 1, 2, \dots, N. \quad (11)$$

The performance characteristic of a dual-blade cluster tool is discussed next. Equations (9) and (10) show the relationship between the number of chambers, the process times, and the robot transfer times in a dual-blade cluster tool. A cursory glance at (7)–(10) show that a dual-blade cluster tool has a FP that is at least as good as a single-blade cluster tool. This is because a dual-blade robot uses the additional blade to introduce a new wafer into a chamber as soon as a processed wafer is withdrawn. With a single-blade cluster tool, the robot will need to travel back and forth twice to replace a processed

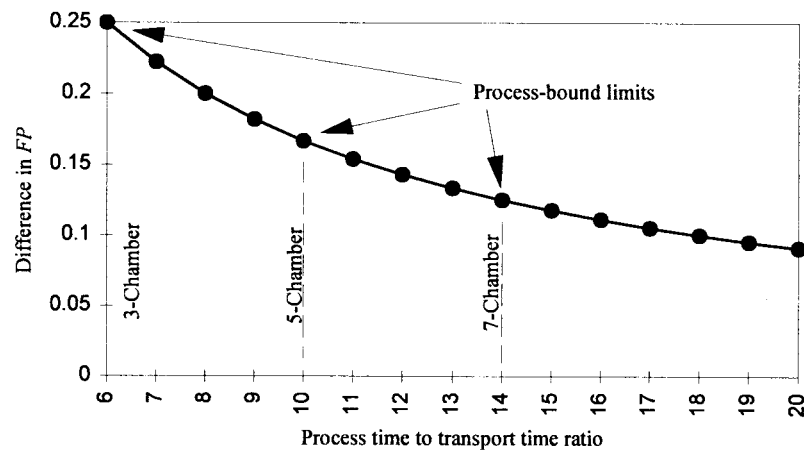


Fig. 7. Difference in FP between dual-blade and single-blade robots in a three-, five-, and seven-chamber cluster tool in the process-bound region (the difference is expressed as a fraction of the dual-blade FP).

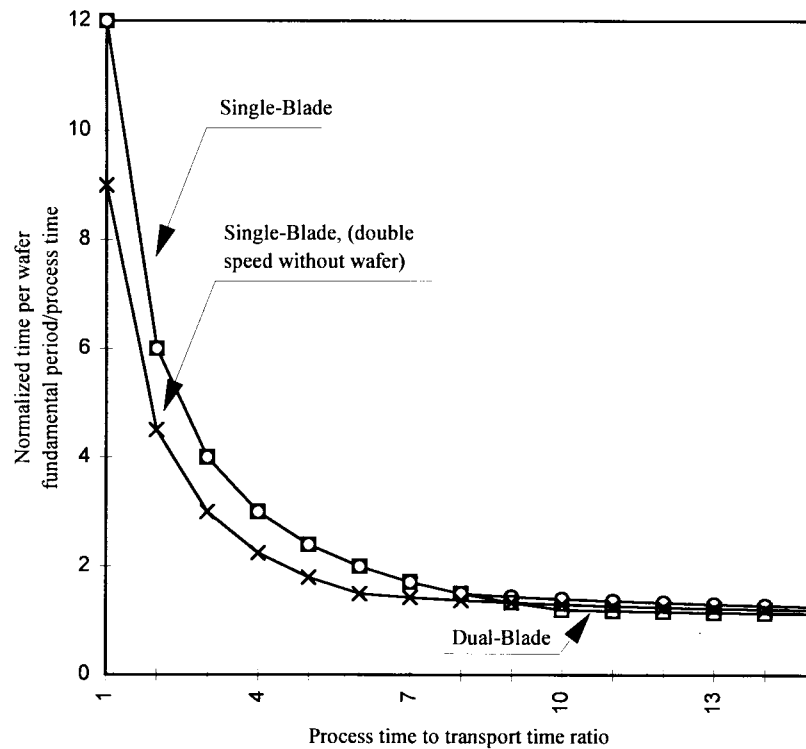


Fig. 8. A comparison of performance of a dual-blade robot with a single-blade robot (with varying speeds) in a five-chamber cluster tool.

wafer with a new wafer in a chamber. This accounts for the difference of $2T$ time units in the process time limits in dual-blade and single-blade cluster tools. The process limit differentiates between the transport-bound and process-bound regions in a cluster tool. Also notice that when a single-blade cluster tool reaches the process-bound region, the robot will need to make additional trips to keep the longest process chamber (P) busy. This additional trip accounts for the $2T$ difference in the FP in (8) and (10) between the dual-blade and single-blade robots. As a consequence, the throughput of a dual-blade cluster tool is considerably higher under process-bound conditions.

In the transport-bound region, an increase in throughput can be achieved only by decreasing the robot transfer times.

Changes in process times, so long as they are within the bounds shown in (11), will not affect the overall throughput of the cluster tool. In the process-bound region, changes in process times and transport times will affect the throughput of cluster tool. The relationship between the different factors is displayed in Figs. 4, 5, and 6, for a three-, five-, and seven-chamber cluster tool, respectively.

In the figures, it is clear that as the number of processing steps increase in the cluster tool, the longer the cluster tool stays in the transport-bound region. However, in comparison to a dual-blade cluster tool, a single-blade tool moves in to the process-bound region earlier (Figs. 4–6). But the FP of the dual-blade cluster tool is nonetheless shorter than a cluster tool with a single-blade

robot of equivalent speed. This is an advantage, since along with shorter FP and, consequently, better throughput, the designer can achieve better performance for shorter process times using a dual-blade instead of a single-blade robot.

In the process-bound region, the dual-blade cluster tool performs better than a single-blade cluster tool. This can be seen in (8) and (10). Fig. 7 shows the difference in the FP between the dual-blade and single-blade configurations, expressed as a fraction of the FP of the dual-blade cluster tool. The larger the difference in the FP, the larger the drop in throughput by using a single-blade instead of a dual-blade robot in the cluster tool.

In Fig. 7, it can be seen that when the process time to transport time ($P:T$) ratio is small, the saving of $2T$ time units in the FP by the use of a dual-blade robot can be a substantial advantage. For example, in the case of a three-chamber cluster tool the difference in the FP can be as much as 25% and, thereby, can result in a 20% drop in throughput by a single-blade cluster tool, when compared with a dual-blade cluster tool. As the $P:T$ ratio increases, the throughput advantage of a dual-blade robot decreases. This is understandable since the process time now predominates the effect on the FP with increasing $P:T$ ratio and, consequently, the type of robot that is used does not have much of an effect on the throughput. Notice also that as the number of chambers increase, the dual-blade robot takes longer to reach the process-bound region. Consequently, the difference in the FP due to the usage of different types of robots in a seven-chamber cluster tool (as the seven-chamber tool reaches the process-bound region) is less, when compared to a three-chamber cluster tool as the three-chamber tool reaches the process-bound region.

Perkinson *et al.* [5] suggest a multiple speed robot to improve the throughput of a single-blade cluster tool. Here, the robot would have different speeds depending on whether or not the robot is carrying a wafer. The robot would typically travel at higher speeds without a wafer than with a wafer. Fig. 8 shows the distinction between the performance curves of the different possibilities against the dual-blade cluster tool. Included in the figure are the performance curves under the following conditions. 1) A single-blade robot with the same speed as that of a dual-blade robot; 2) a single-blade robot with double the speed of a dual-blade robot, when the single-blade robot is traveling with a wafer; and 3) a dual-blade robot. It can be seen that in the transport-bound region, by doubling the speed of transfer when the robot is not transferring a wafer, a single-blade robot can achieve better performance than a dual-blade robot. However, in the process-bound region, the dual-blade robot still performs better than the single-blade robot. In fact looking at (8) and (10), it is clear that a single-blade has to double the overall speed (with or without a wafer) to achieve performances similar to a dual-blade robot. Note that the maximum allowable speed of the robot is determined by the wafer-holding mechanism. A relative motion between the wafer-holding mechanism and the wafer can introduce particle contamination and, consequent, defects in the wafer.

V. CONCLUSION

An analysis of steady-state throughput for a dual-blade single-wafer cluster tool is presented. The analysis shows that a dual-blade robot performs at least as well as a single-blade robot in the case of serial-processing. In the transport-bound case, the throughput of the cluster tool is the same using either a dual and single-blade robot. In the process-bound region, the cluster tool with a dual-blade performs better than with a single-blade robot. The difference in performance tends to decrease as the $P:T$ ratio and/or the number of chambers in the cluster tool increases. In the case of a three-chamber cluster tool the difference in the FP of the cluster tool can be as much as 25%, compared with the FP of the dual-blade cluster tool. This can translate to a substantial difference in the throughput. In the case of five-chamber or seven-chamber cluster tools, this can be as much as 16.6% and 12.5%, respectively.

Significantly, it was noted that the cluster tool moves from a transport-bound region to a process-bound region more quickly with a single-blade robot, than a dual-blade robot of equivalent speed. This implies that one can achieve better throughput at even lower process times with a dual-blade robot, than with a single-blade robot in the cluster tool. Finally, it was demonstrated that a single-blade robot would need to be at least twice as fast as a dual-blade robot to achieve similar throughput in the cluster tool in the process-bound region.

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Srilakshmi Venkatesh (S'94–A'96) received the B.E. degree in mechanical engineering from Bangalore University, Bangalore, India, and the Ph.D. degree in industrial engineering from Texas A&M University, College Station, in 1996.

Since then, he has been an analyst with Global Products Management for the Metal Deposition Products Business group, Applied Materials, Santa Clara, CA. He has been involved in the management and deployment of operations research and computer science technologies and tools to improve a

variety of products business metrics, and has a patent pending in cluster tool scheduling/control.

Dr. Venkatesh is a member of INFORMS, IIE, and SME.



Rob Davenport received the B.S. degree in mechanical engineering from Oklahoma State University, Stillwater.

He has served as an Officer in the U.S. Navy as a Nuclear Engineer aboard an aircraft carrier. He is currently the Director of Global Product Management for the PVD Products Business Group, Applied Materials, Santa Clara, CA. Since joining Applied Materials in 1989, he has been the Endura HP and the Endura VHP program Manager, and the Product Manager for MCA E-Chuck. He has worked on various development programs and market introductions, and holds three patents on E-chuck and wafer clamping mechanisms.

Pattie Foxhoven received the B.S. degree in computer science and statistics from Roanoke College, Salem, VA, in 1987.

Since joining Applied Materials, Santa Clara, CA, in 1987, she has worked in software development, marketing, and product management. She is currently the Software and Factory Automation Product Manager for the Applied Materials Metallization Division. Her current interests are in semiconductor fab automation and productivity improvement.

Jaim Nulman (M'83–SM'90) received the B.S. degree in electrical engineering from the Technion—Israel Institute of Technology, Haifa, and the M.S. and Ph.D. degrees in electrical engineering from Cornell University, Ithaca, NY.

He continued for two years as a Research Associate with Cornell's School of Electrical Engineering and the National Submicron Facility, where he worked on advanced technologies for submicron silicon and III–V devices. While at Cornell, he pioneered the use of rapid thermal processing for oxidation, nitration of oxides and formation of titanium silicide for <0.5 μm applications. He has co-invented several techniques for compensation of wafer emissivity effects in pyrometric temperature monitoring. He is the Managing Director and Global Product Manager for PVD Products Business Group, Applied Materials, Santa Clara, CA. Prior to this he was Product Manager for PVD and a manager of Process Technology Integration for the Endura PVD System. He joined Applied Materials in 1989. Prior to coming to Applied Materials, he worked with AG Associate for four years, where he was the Technology Manager responsible for new process applications and RTP technology development. He has presented numerous invited and contributed papers on rapid thermal processing and advanced devices, and has served as instructor for NATO's Advanced Studies Institute, as well as at the University of California, Berkeley Extension.

Dr. Nulman is a member of ECS, APS, and MRS.